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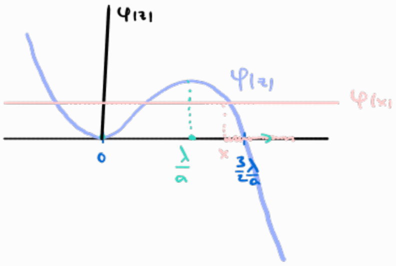

$$\Longrightarrow \quad u(t) \geq x > \frac{\lambda}{a} \quad \forall t \in I = [0, T_{max})$$

$$-u''(t) = \lambda u(t) - a u^2(t) = \underset{\overset{\wedge}{0}}{(\lambda - a u(t))}\, \underset{\overset{\vee}{0}}{u(t)} < 0 \Longrightarrow u''(t) > 0 \quad \forall t \in I \Longrightarrow (u')'(t) > 0 \Longrightarrow u'(t) \text{ creciente}$$

$$u(t) > \frac{\lambda}{a} \quad , \quad \lambda - a u(t) < 0$$

Como $u'(0) = 0$, $u'(t)$ crece $\Longrightarrow \underset{(v(t))}{u'(t) > 0}$ $\forall t \in (0, T_{max})$ y $u(t)$ creciente

$$\frac{v^2(t)}{2} = \frac{a}{3}\left(u^3(t) - x^3\right) - \frac{\lambda}{2}\left(u^2(t) - x^2\right)$$

$$v^2(t) = \frac{2a}{3}\left(u^3(t) - x^3\right) - \lambda\left(u^2(t) - x^2\right)$$

$$v(t) = +\sqrt{\frac{2a}{3}\left(u^3(t) - x^3\right) - \lambda\left(u^2(t) - x^2\right)}$$

$\Downarrow$ ya sabemos que es positiva

• Vamos a ver que explota cuando $x > \frac{\lambda}{a}$

$$t = \int_0^t 1 = \int_0^t \frac{u'(s)\, ds}{\sqrt{\frac{2a}{3}\left(u^3(t) - x^3\right) - \lambda\left(u^2(s) - x^2\right)}} \underset{u(s)=z}{\overset{=}{\Downarrow}} \int_x^{u(t)} \frac{dz}{\sqrt{\frac{2a}{3}\left(z^3 - x^3\right) - \lambda\left(z^2 - x^2\right)}} \underset{z = x\theta}{\overset{=}{\Downarrow}} \int_1^{u(t)/x} \frac{x\, d\theta}{\sqrt{\frac{2a}{3} x^3\left(\theta^3 - 1\right) - \lambda x^2\left(\theta^2 - 1\right)}} = \int_1^{u(t)/x} \frac{d\theta}{\sqrt{\frac{2a}{3} x\left(\theta^3 - 1\right) - \lambda\left(\theta^2 - 1\right)}} < \quad \forall t \in [0, T_{max})$$

• Vamos a ver que es explosiva

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$$< \; j \equiv \int_1^{\infty} \frac{d\theta}{\sqrt{\frac{2a}{3} x\left(\theta^3 - 1\right) - \lambda\left(\theta^2 - 1\right)}} < \infty \Longrightarrow t < j \quad \forall t \in [0, T_{max}) \Longrightarrow T_{max} \leq j < \infty$$

Por lo tanto sabemos que $\overline{\lim_{t \uparrow T_{max}}} \left(|u(t)| + |u'(t)|\right) = +\infty$

Como $u(t)$ y $\underset{v(t)}{\overset{\shortparallel}{u'(t)}}$ son crecientes $\Longrightarrow \lim_{t \uparrow T_{max}} \left(u(t) + u'(t)\right) = +\infty \Longrightarrow \lim_{t \uparrow T_{max}} u(t) = +\infty$, $\lim_{t \uparrow T_{max}} u'(t) = +\infty$

$$T_{max} = \int_1^{\infty} \frac{d\theta}{\sqrt{\frac{2a}{3} x\left(\theta^3 - 1\right) - \lambda\left(\theta^2 - 1\right)}}$$

- Vamos a estudiar que efectivamente esa integral es convergente

$$T_{max}(x) = \int_1^{\infty} \frac{d\theta}{\sqrt{\frac{2a}{3}x(\theta^3-1) - \lambda(\theta^2-1)}} < \infty \quad , \text{ con } x > \frac{\lambda}{a} \qquad \left(\frac{\lambda}{a}, +\infty\right)$$

Si x crece, $T_{max}(x)$ decrece : $\lim\limits_{x \uparrow \infty} T_{max}(x) = 0$

$$\int_1^{\infty} = \int_1^{M} + \int_M^{\infty} \qquad M > 1 \quad . \quad f(\theta) = \frac{2ax}{3}(\theta^3-1) - \lambda(\theta^2-1) = (\theta-1)\Big[\underbrace{\frac{2ax}{3}(\theta^2+\theta+1) - \lambda(\theta+1)}_{\overset{\parallel}{g(\theta)}}\Big] .$$

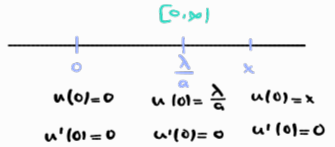
- $$\int_1^M \frac{d\theta}{\sqrt{f(\theta)}} = \int_1^M \frac{d\theta}{\sqrt{(\theta-1)\,g(\theta)}} \leq \qquad g'(\theta) = \frac{2ax}{3}(2\theta+1) - \lambda > \frac{2ax}{3}\cdot 3 - \lambda = 2ax - \lambda > 2a\frac{\lambda}{a} - \lambda = \lambda > 0 \implies g \text{ estrictamente creciente}$$

$$\implies g(\theta) \geq g(1) = \frac{2ax}{3}\cdot 3 - 2\lambda = 2(ax-\lambda) > 0$$

$$\leq \int_1^M \frac{d\theta}{\sqrt{(\theta-1)\,2(ax-\lambda)}} = \frac{1}{\sqrt{2(ax-\lambda)}} \cdot \int_1^M (\theta-1)^{-\frac{1}{2}} d\theta = \frac{2}{\sqrt{2(ax-\lambda)}} \cdot \Big[\sqrt{\theta-1}\Big]_{1=\theta}^{M=\theta} = \frac{2\sqrt{M-1}}{\sqrt{2(ax-\lambda)}} \longrightarrow 0 \text{ si } x \uparrow \infty$$

- $$\int_M^{\infty} \frac{d\theta}{\sqrt{\frac{2ax}{3}(\theta^3-1) - \lambda(\theta-1)}} = \int_M^{\infty} \frac{\sqrt{\frac{2ax}{3}\theta^3}}{\underbrace{\sqrt{\frac{2ax}{3}(\theta^3-1) - \lambda(\theta^2-1)}}_{1 \text{ cuando } \theta \uparrow \infty \quad \leq 2 \text{ si } x \geq M}} \cdot \frac{d\theta}{\sqrt{\frac{2ax}{3}\theta^3}} \leq 2 \cdot \int_M^{\infty} \frac{d\theta}{\sqrt{\frac{2ax}{3}\theta^3}} = \sqrt{\frac{3}{2ax}} \cdot 2 \cdot \int_M^{\infty} \theta^{-\frac{3}{2}} d\theta = -4\cdot\sqrt{\frac{3}{2ax}} \cdot \left[\frac{1}{\sqrt{\theta}}\right]_{\theta=M}^{\theta=\infty} =$$

$$= 4 \cdot \sqrt{\frac{3}{2ax}} \cdot \frac{1}{\sqrt{M}} \longrightarrow 0 \text{ si } x \uparrow \infty$$

- 

Comprobemos por dependencia Lipschitziana que $T_{max}(x) \to \infty$ si $x \downarrow \frac{\lambda}{a}$

Se puede ver que esta parte de la integral diverge:

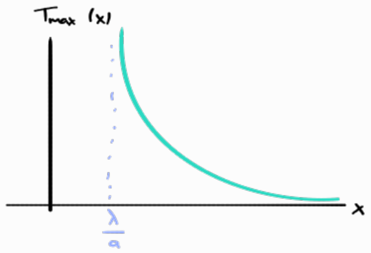
$$\int_1^M \frac{d\theta}{(\theta-1)\,g(\theta)} \quad , \quad g(\theta) = \frac{2ax}{3}(\theta^2+\theta+1) - \lambda(\theta+1)$$

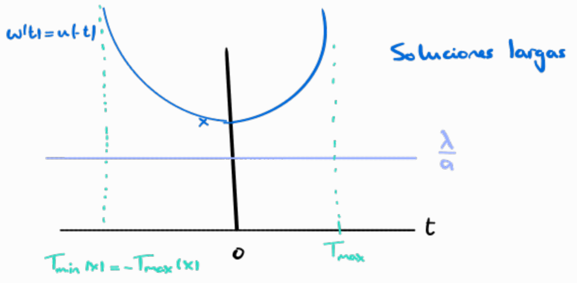
$$\underset{\substack{\downarrow \\ x-\frac{\lambda}{a}}}{g(\theta) =} \frac{2\lambda}{3}(\theta^2+\theta+1) - \lambda(\theta+1) \quad , \quad g(1) = 0$$

$$\begin{array}{c|ccc} & \frac{2\lambda}{3} & \overset{\overset{-\frac{\lambda}{3}}{\parallel}}{\frac{2\lambda}{3}-\lambda} & \overset{\overset{-\frac{\lambda}{3}}{\parallel}}{\frac{2\lambda}{3}-\lambda} \\ 1 & & \frac{2\lambda}{3} & \frac{\lambda}{3} \\ \hline & \frac{2\lambda}{3} & \frac{\lambda}{3} & \boxed{0} \end{array}$$

$$g(\theta) = (\theta-1)\left(\frac{2\lambda}{3}\theta + \frac{\lambda}{3}\right)$$

$$\int_1^M \frac{d\theta}{\sqrt{(\theta-1)\,g(\theta)}} = \int_1^M \frac{d\theta}{\sqrt{(\theta-1)^2\left(\frac{2\lambda}{3}\theta + \frac{\lambda}{3}\right)}} = \int_1^M \frac{d\theta}{(\theta-1)\cdot\sqrt{\frac{\lambda}{3}(2\theta+1)}} \geq \frac{1}{\sqrt{\frac{\lambda}{3}(2M+1)}} \cdot \int_1^M \frac{d\theta}{\theta-1} = +\infty$$





- Hasta ahora hemos estado estudiando ecuaciones de este tipo $\begin{cases} u'' = f(t, u, u') \\ u(t_0) = u_0 \\ u'(t_0) = v_0 \end{cases}$